

Complications of Trans-oral Endoscopic Thyroidectomy Vestibular Approach: A Systematic Review.

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This review focuses on complications linked to trans-oral endoscopic thyroidectomy via vestibular approach (TOETVA) and aimed to elucidate the procedure's initial safety profile. According to the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA), Pubmed, Embase, and the Cochrane databases were screened till May 2021. Twenty-eight articles, nine cohorts and nineteen case series, met the inclusion criteria. Procedure-related complications were analyzed, the most important being hypoparathyroidism: transient (range=0.94-22.2%), permanent (range=1.33-2.22%), and recurrent laryngeal nerve injury: transient (range=1.9-8.8%) and permanent (range=0.59-1.42%). Surgical trauma related complications, the most prevalent being seroma, emphysema, and hematoma accounted for 2.91%. Null mortality was reported. Although current evolving experience indicates that TOETVA is safe and linked to acceptable complication rates, the method needs to be compared with the gold standard of traditional thyroidectomy in the context of sufficiently numbered cohorts and ultimately randomized controlled trials.

Imaging Findings of Liposuction with an Emphasis on Postsurgical Complications.

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Liposuction is one of the most frequently performed cosmetic surgeries worldwide for reshaping the body contour. Although liposuction is minimally invasive and relatively safe, it is a surgical procedure, and it carries the risk of major and minor complications. These complications vary from postoperative nausea to life-threatening events. Common complications include infection, abdominal wall injury, bowel herniation, bleeding, haematoma, seroma, and lymphoedema. Life-threatening complications such as necrotizing fasciitis, deep vein thrombosis, and pulmonary embolism have also been reported. In this paper, we provide a brief introduction to liposuction with the related anatomy and present computed tomography and ultrasonography findings of a wide spectrum of postoperative complications associated with liposuction.

Seroma Formation in Pre-pectoral Implant-Based ADM Assisted Breast Reconstruction: A Comprehensive Review of Current Literature.

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In the field of implant-based breast reconstruction (IBBR), the most recent and successful progress has been the evolution of the prepectoral approach through the use of acellular dermal matrix (ADM). ADM-assisted breast reconstruction is now gaining a foothold as standard practice, but its advantages are often clouded due to discouraging studies reporting increased seroma formation. The origin of the serum accumulation still remains unclear, but it has always been the most frequent early complication in breast reconstruction, thus proving to be crucial to address since it can lead to further complications. Using a standardized approach to obtain high-quality scientific evidence, the aim of this review is therefore to investigate the occurrence of seroma in breast implant-based reconstructive surgery and its possible relation with matrices. We conducted the review investigating only pre-pectoral implant positioning and

one specific ADM (Braxon®) which is the one who has the highest number of cases in literature. Interestingly, ADM does not appear to be causative of seroma formation, but a surgery-related origin is discussed. In this setting, a series of rigorous guidelines have therefore been identified and analyzed to allow not only the treatment, but also the prevention of seroma, thus leading to a reduction in the incidence of this frequent problem.

Imaging of Abdominal Wall Masses, Masslike Lesions, and Diffuse Processes.

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Abdominal wall masses, masslike lesions, and diffuse processes are common and often incidental findings at cross-sectional imaging. Distinguishing among these types of masses on the basis of imaging features alone can be challenging. The authors present a diagnostic algorithm that may help in distinguishing different types of abdominal wall masses accurately. Hernias may mimic discrete masses at clinical examination, and imaging is often ordered for evaluation of a possible abdominal wall mass. Once a discrete mass is confirmed to be present, the next step is to determine if it is a fat-containing, cystic, or solid mass. The most common fat-containing masses are lipomas. Fluid or cystic masses include postoperative abscesses, seromas, and rectus sheath hematomas. Solid masses are the most common abdominal wall masses and include desmoid tumors, sarcomas, endometriomas, and metastases. Multiple masses and other diffuse abdominal wall processes are often manifestations of an underlying condition or insult. The most frequently found diffuse processes are multiple injection granulomas from administration of subcutaneous medication. This article offers an algorithmic approach to characterizing abdominal wall masses on the basis of their composition and reviews abdominal wall diffuse processes. Online supplemental material is available for this article. ©RSNA, 2020.

Collaboration in developing a protocol for nurse-led seroma aspiration.

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A seroma is a serous fluid collection which may develop in the space between the chest wall and skin flaps following breast cancer surgery with axillary lymph node dissection. Seroma formation following mastectomy is most common. This article aims to produce a protocol for nurse-led seroma aspiration in collaboration with the breast multidisciplinary team and describe the four main steps involved in the development. It was important to develop a protocol for nurse-led seroma aspiration to enable the breast cancer clinical nurse specialist (CNS) to practice confidently and safely in an extended area of practice. The protocol will assist the breast cancer CNS by providing clear and specific guidance. CNSs considering developing a protocol for seroma aspiration should involve all relevant members of the breast multidisciplinary team. Breast cancer CNSs must be trained in seroma aspiration and undergo clinical supervision. Only when deemed competent, should nurse specialists provide a seroma aspiration service independently. Patient satisfaction with the nurse-led seroma aspiration service should be evaluated by clinical audit at an early stage.

Is breast seroma after tumour resection associated with patient-reported breast

appearance change following radiotherapy? Results from the IMPORT HIGH (CRUK/06/003) trial.

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Seroma describes a collection of serous fluid within a cavity, occurring following surgery. Seroma is associated with normal tissue effects (NTE) following breast radiotherapy, as reported by clinicians and on photographs. This study investigates the association between seroma and the NTE breast appearance change collected using patient-reported outcome measures (PROMs) in IMPORT HIGH, as well as investigating the association between breast appearance change and patient/tumour/treatment factors. Case-control methodology was used for seroma analysis within IMPORT HIGH. Cases were patients reporting moderate/marked breast appearance change and controls reported none/mild changes at year-3. One control was selected at random for each case. Seromas were graded as not visible/subtle or visible/highly visible on CT radiotherapy planning scans. Logistic regression tested associations, adjusting for patient/tumour/treatment factors. 1078/1149 patients consented to PROMs, of whom 836 (78%) reported whether they had 3-year breast appearance change; 231 cases and 231 controls were identified. 304/462 (66%) patients received chemotherapy. Seroma prevalence was 20% (41/202) in cases and 16% (32/205) in controls, and less frequent in patients receiving adjuvant chemotherapy [10% (24/246) compared with 29% (40/138) without]. Visible seroma was not significantly associated with breast appearance change [OR 1.38 (95%CI 0.83-2.29), $p = 0.219$]. Larger tumour size, haematoma, current smoking and body image concerns at baseline were independent risk factors. Seroma was not associated with patient-reported breast appearance change, but haematoma and smoking were significant risk factors. Lack of association may be related to lower prevalence of seroma compared with previous reports, perhaps reflecting patients receiving adjuvant chemotherapy in whom seroma resolves prior to radiotherapy.

Seroma formation after breast cancer surgery: what we have learned in the last two decades.

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Formation of a seroma most frequently occurs after mastectomy and axillary surgery. Prolonged drainage is troublesome as it increases the risk for infection and can significantly delay adjuvant therapy. Seroma has been defined as serous fluid collection under the skin flaps or in the axillary dead space following mastectomy and/or axillary dissection. Because the true etiology of a seroma is unknown, a multifactorial-causation hypothesis has been accepted. Surgical factors include technique, extent of dissection and the surgical devices used for dissection. Obliteration of dead space with various flap fixation techniques, use of sclerosants, fibrin glue and sealants, octreotide, and pressure garments have been attempted with conflicting results and none have been consistent. Early movement of the shoulder during the postoperative period may increase the formation of seroma, although delayed physiotherapy decreases the formation of seroma. A detailed analysis of the use of drains showed that use of single or multiple drains, early or late removal, and drains with or without suction are not significantly different for the incidence of seroma. Although there is evidence for reduced seroma formation after early drain removal, very early removal within 24 hours seems to increase formation of seroma. No patient or tumor factors seem to affect seroma formation except body mass index and body weight. Consensus is lacking among studies/trials with different groups producing conflicting evidence. Besides a few established factors such as body mass index, the use of electrocautery for dissection, early drain removal, low

vacuum drains, obliteration of dead space, and delayed shoulder physiotherapy, most of the hypothesized causes have not been demonstrated consistently. Thus, seroma remains a threat to both the patient and surgeon. Recurrent transcutaneous aspiration remains the only successful management.

Predisposing factors for seroma formation in patients undergoing thyroidectomy: Cross-sectional study.

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Seroma is defined as collection of fluid within the surgical site during postoperative period that causes several complications. Recognition of predisposing risk factors can lead to avoid seroma formation after thyroidectomy. A cross-sectional study was carried out during a 3-year period and 678 patients were enrolled the study. We recorded demographic data, past medical history and the type of thyroidectomy were for all patients. We measured total and ionized serum calcium and albumin level in all patients before surgery and a day after it. All patients underwent total or subtotal thyroidectomy and if needed central neck dissection was performed subsequently. Patients underwent serial aspiration if they had seroma formation. The overall post-thyroidectomy seroma incidence was 2.2%. There was no statistically significant correlation while evaluating gender, age and body mass index with post-operative seroma formation. However, seroma formation was significantly higher in patients underwent total thyroidectomy ($P = 0.041$). The results of postoperative laboratory tests showed a significant lower level of ionized calcium in patients with seroma formation ($P < 0.0001$). Logistic regression showed statistically significant value for variables including age, BMI and decreased ionized calcium level, in developing of seroma. We showed that Seroma formation was lower during thyroidectomy via electrical vessel sealing system in comparison with previous studies. In our study, older age, greater body mass index and decreased ionized calcium level were predictors of seroma formation.

Seroma fluid subsequent to axillary lymph node dissection for breast cancer derives from an accumulation of afferent lymph.

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Seroma is a frequent complication of breast cancer surgery, the etiology of which remains indefinite. It represents a subcutaneous accumulation of fluid frequently reported after surgical procedures such as axillary lymph node dissection. Despite previous studies have associated seroma fluid to an inflammatory exudate, the surgical removal of draining lymph nodes may indicate that seroma might not represent a mere exudate but rather an accrual of lymph drained from tributary tissues. To verify this hypothesis, seromas were collected at different intervals of time in patients operated upon for axillary lymph node removal. Fluids were analyzed in details by flow cytometry and biochemical assays for their cellular content and for their molecular features and relevant cytokine content. Lymphocytes and other peculiar blood mononuclear cells were present, while erythrocytes, platelets and granulocytes were absent or extremely rare. The protein concentration resulted lower (median 64%) than in peripheral blood. However, specific proteins related to locoregional tissues resulted highly concentrated (e.g. up to 500% for ferritin and 300% for lactate dehydrogenase and exclusive presence of interleukin-6) whereas all enzymes and proteins synthesized in the liver or other organs (e.g. alkaline phosphatase, ALT, gammaGT, prealbumin, transferrin, ceruloplasmin, C3 and C4, alpha2 macroglobulin from liver;

apolipoproteins from liver and gut; amylase and lipase from pancreas) were represented in reduced concentrations, thus ruling out that seroma proteins derive directly from blood serum. As a whole, this comprehensive cytological and molecular analysis provided evidences that seroma is constituted by serum ultrafiltrated-derived extracellular fluid of regions located upstream of removed lymph nodes. This fluid is then enriched by proteins and cells collected in the drained regions. Remarkably, seroma fluids collected in the same patient at different time points (up to 50 days following surgery) displayed similar biochemical features, clearly indicating that fluid composition was not significantly affected by post-surgical locoregional flogosis. Finally, the period of seroma formation indicates that lymph accumulates in the axillary region during the interval of time needed for afferent lymphatic vessels to re-anastomose with the efferent ducts. Therefore, seroma fluid represents a font of biological material suitable for investigating the biology of breast cancer, healing tissues and lymph.

[Seroma: a complication of systemic-pulmonary shunt with a tubular prosthesis].

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The accumulation of clear fluid leaking from a tubular prosthesis and confined within a fibrous pseudomembrane has been defined as a "seroma". Both the literature and our experience on this complication have been reviewed. Fifty-three cases of seroma have been reported after implantation of a tubular prosthesis. In our experience, from June 1982 to September 1988 we observed 11 cases (11/108 = 10.1%) of seroma complicating a modified Blalock-Taussig procedure. The mean age of these patients was 2 years 2 months. The diameter of the implanted tubular prostheses was always 5 mm. Among these cases, we report the details of a child, 1 year 10 months old, in whom the seroma significantly complicated the surgical history. According to the data derived from the literature and from our experience, we derived the following risk factors for the occurrence of seroma: age of the patient, size of the prosthesis, surgical manipulations, arterial-venous gradient, transprosthetic flow.
